

ECE-111 Advanced Digital Design Projects

Homework-3a:

Develop SystemVerilog RTL model for 4-bit Johnson Counter and perform following mentioned :

- Synthesize Johnson counter
- Run simulation using Johnson counter testbench provided
- Review synthesis results (resource usage and RTL netlist/schematic)
 - Note : Explanation of resource usage in report is not mandatory to provide.
- Review input and output signals in simulation waveform.

About Johnson Counter :

- Similar to Straight Ring counter with below mentioned difference
- Johnson counter connects the complement of the output of the last shift register to the input of the first register and circulates a stream of ones followed by zeros around the ring
- Johnson Counter has $2N$ states (if N is 4, then total states for counting is 8. And 4-bit pattern will repeat after 8 cycles).

4-bit Johnson Counter with initial load count value = 4'b0000

Johnson Counter					
Clock Cycle	State	count[3]	count[2]	count[1]	count[0]
1	0	0	0	0	0
2	1	1	0	0	0
3	2	1	1	0	0
4	3	1	1	1	0
5	4	1	1	1	1
6	5	0	1	1	1
7	6	0	0	1	1
8	7	0	0	0	1
9	0	0	0	0	0

4-bit
pattern
repeats
every 8
cycles

HW-3b:

Report the output sequence if we preload with one of the 8 remaining states not covered by preloading with 0000.

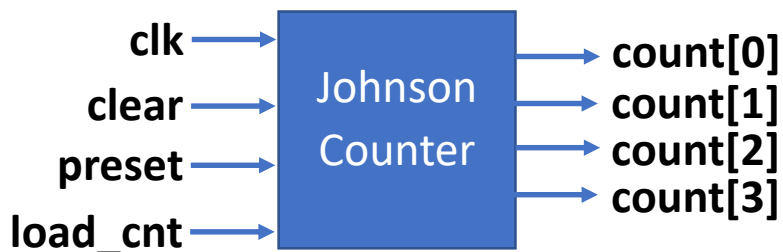
Homework Submission :

Submit report (PDF file) which should include:

- Snippet of SystemVerilog design code
- RTL netlist viewer schematic and Synthesis resource usage
 - Optional: Resource usage explanation and post mapping schematic
- Simulation snapshot and explain simulation result to confirm it works as a Johnson counter
- For part 3-b, the sequence of count[3:0] values, in binary.

Note:

- When creating RTL model for Johnson Counter, name SystemVerilog Module name as : **johnson_counter**
- Assume below mentioned Primary Ports for Johnson Counter
 - Input clk (clock)
 - Input clear (asynchronous reset / negedge signal)
 - Input preset (synchronous and active low signal)
 - Input load_cnt (value gets loaded when !preset)
 - Output count (output count value)



Lab3.zip folder posted on Piazza and Canvas has **johnson_counter_testbench.sv** testbench code to simulate with johnson counter SystemVerilog module.

Note :

- 3-a: Assume initial load_cnt signal value is set to 4'b0000 in testbench code provided.
- 3-b: Students update johnson_counter_testbench.sv code for initial load_cnt value !0.